Last week:

- Matchings
- Edge- and vertex-covers
- Edge- and vertex- independence and covering numbers $\alpha, \alpha', \beta, \beta'$.

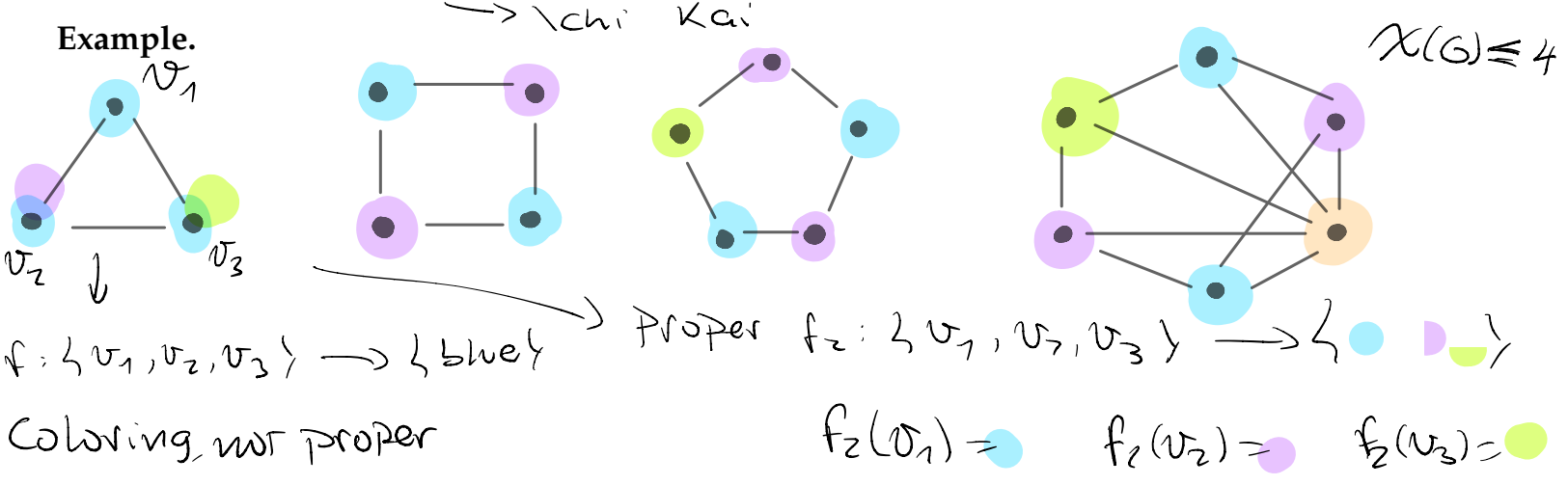
This week:

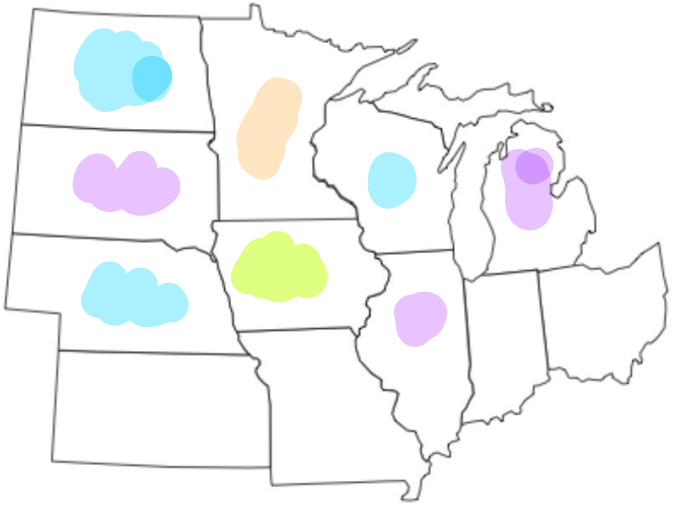
- Vertex coloring
- Chromatic numbers

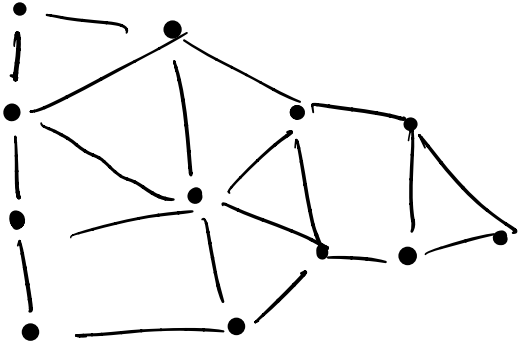
Definition. *Let G be a graph.*

- *A **coloring** of G is a function $f\colon V(G) \to C$ where C is a non-empty set.*
- *A **proper coloring** of G is a coloring $f\colon V(G) \to C$ such that if $uv \in E(G)$, then $f(u) \neq f(v)$. In this case C is called the **palette** and the elements in C are **colors**.*
- *For an integer $t \geqslant 1$, a **t-coloring** of G is a proper coloring $f\colon V(G) \to C$ such that $|C| \leqslant t$.*
- *The **chromatic number** $\chi(G)$ is the smallest integer t such that G has a t-coloring.*

Example.









Quick observations:

(a) If G is an n-vertex graph, then $\chi(G) \leq n$. Also, $\chi(G) = n$ if and only if $G \cong K_n$.

$\chi(G) \leq n$: We have a proper coloring using n colors.

If $G \cong K_n$, we need n colors. If $G \ncong K_n$, Take u, v such $uv \notin E(G)$

(b) $\chi(G) = 1$ if and only if G has no edges.

$\chi(G) = 1 \Rightarrow$ no edges ✓

no edges $\Rightarrow$ can use one color.

Then u, v can use the same color
Take $n-2$ new colors for the $n-2$ remaining vertices
This gives $(n-1)$-coloring

(c) If H is a subgraph of G, then $\chi(H) \leq \chi(G)$.

If $f: V(G) \to C$ is minimum proper coloring, $|C| = \chi(G)$

Then $f: V(H) \to C$ is a proper coloring, so $\chi(H) \leq |C| = \chi(G)$.

Definition. *The **clique-number** $\omega(G)$ of a graph G is the largest size of a clique contained in G.*

Clique $\equiv$ complete graph.

(d) $\chi(G) \geq \omega(G)$

By def^n, $K_{\omega(G)} \subseteq G$, by (c) $\chi(K_{\omega(G)}) \leq \chi(G)$

$\omega(G)$ by (a).

$\{v \in V(G) : f(v) = c\}$

Definition. *A **color class** of a proper coloring* $f: V(G) \to C$ *is the set* $f^{-1}(c)$ *for any* $c \in C$.

Proposition. *Proper colorings of* G *using* t*-colors correspond to partitions of* $V(G)$ *into* t *independent sets. In particular,* $\chi(G)$ *is the smallest integer* t *for which* G *is* t*-partite.*

Proof: Take a proper t-coloring $f: V(G) \longrightarrow C = \{c_1, \dots, c_t\}$

For each $1 \le i \le t$, Take

$$A_i = f^{-1}(c_i) = \{v \in V(G) : f(v) = c_i\}$$

Then $V(G) = A_1 \cup A_2 \cup \dots \cup A_t$ and each A_i is independent because if $u, v \in A_i$, Then $f(u) = f(v)$, Thus $uv \notin E(G)$

Conversely, if $V(G) = A_1 \cup \dots \cup A_t$ is a t-partition

Define $f: V(G) \longrightarrow \{1, \dots, t\}$, by $f(v) = i$ if $v \in A_i$

This is a proper coloring: if $uv \in E(G)$, Then u, v belong To different parts A_i (because independent) ☒

Theorem (10.5). *If* G *is an* n*-vertex graph, then*

$\chi(G) \geq \frac{n}{\alpha(G)}$ → Independence number

Proof: By the previous prop, we have

$V(G) = A_1 \sqcup \cdots \sqcup A_{\chi(G)}$ $\chi(G)$-partition

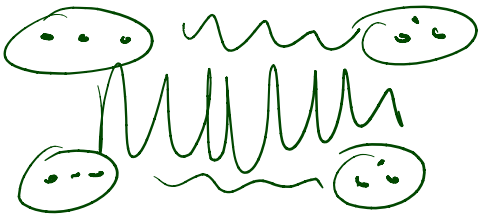
Since each A_i is independent, $|A_i| \leq \alpha(G)$.

Then $n = |V(G)| = \sum_{i=1}^{\chi(G)} |A_i| \leq \sum_{i=1}^{\chi(G)} \alpha(G) = \alpha(G)\chi(G)$.

i.e.: $n \leq \chi(G)\alpha(G)$. ∎

$\alpha(G) \geq \frac{n}{\chi(G)}$

Example ($\chi(G) = \omega(G)$ and $\chi(G) = \frac{n}{\alpha(G)}$).



$G = K_{3,3,3,3}$ complete 4-partite parts of size 3

$n = 12$

$\omega(G) = 4$, $\chi(G) = 4$

Definition. *For a graph* G*, the* ***degeneracy*** *of* G *is*

$$d(G) = \max\{\delta(H) : H \text{ is a subgraph of } G\} \leq \Delta(G)$$

Theorem (10.7, 10.9). *For any graph* G*, we have* $\chi(G) \leq 1 + d(G)$*. In particular,* $\chi(G) \leq 1 + \Delta(G)$.

Proof: Induct on $n = |V(G)|$. $G = \bullet$

Base case $n=1$, Then $d(G) = 0$, and $\chi(G) = 1$ (no edges)

Consider $n \geq 2$. Take $v \in V(G)$ for which $\deg_G v = \delta(G)$

Note: $\deg_G v = \delta(G) \leq d(G)$ (G is a subgraph of G)

Take $H = G - v$. Since H is a subgraph of G, we have $d(H) \leq d(G)$

By induction: $\chi(H) \leq 1 + d(H) \leq 1 + d(G)$

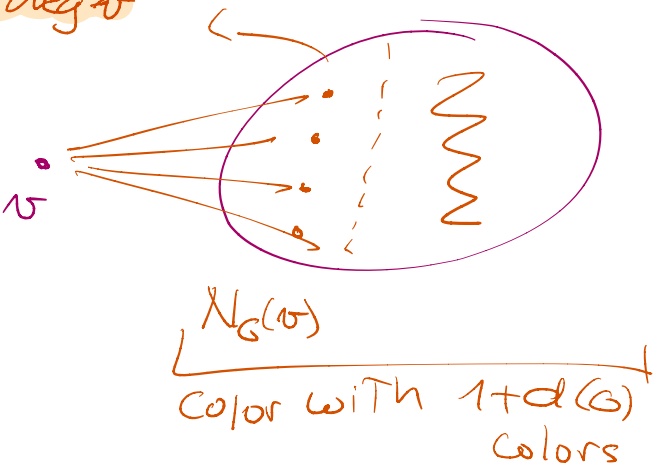
Thus, there is a proper coloring $f: V(H) \to C$ with $|C| = 1 + d(G)$

Next: Extend f to a proper coloring $\tilde{f}: V(G) \to C$

Since $|C| = d(G) + 1$, and $\deg_G v \le d(G)$, There is some $c \in C$ which is unused by $N_G(v)$. Define $\tilde{f}: V(G) \to C$

$$\tilde{f}(w) = \begin{cases} f(w) & \text{if } w \neq v \\ c & \text{if } w = v \end{cases}$$

$d(G) \ge \deg v$



Only need to show: if $u \in N_G(v)$, Then $\tilde{f}(u) \neq \tilde{f}(v)$

This is clear because $\tilde{f}(u) = f(u)$, $\tilde{f}(v) = c$

and c is unused by elements in $N_G(v)$

Thus $\tilde{f}$ is a proper coloring, Then $\chi(G) \le |C| = 1 + d(G)$.

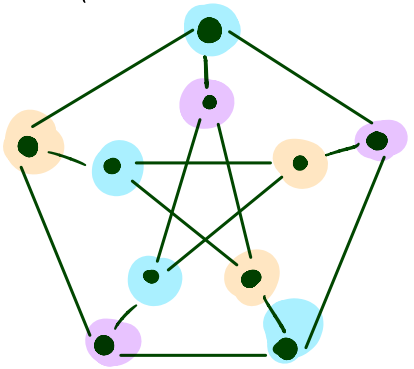


Theorem (Brook's, 10.8). *For a connected graph* G *that is not an odd cycle or a clique, we have* $\chi(G) \leqslant \Delta(G)$.

Example.

P = Petersen

$\chi(P) = ?$



$\chi(P) \geqslant \chi(C_5) = 3$

$\Delta(P) = \delta(P) = 3$

Thus $\chi(P) \leq 3$ (Brook's)